Claude Code Telemetry Analytics Platform

Bojana Cekić

Transforming developer telemetry into actionable insights

- Built an end-to-end analytics platform for Claude Code telemetry data
- The system ingests raw event streams and user metadata
- Processes telemetry to discover developer usage patterns
- Presents insights through an interactive dashboard

System Architecture

Raw Telemetry Logs

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Data Ingestion

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SQLite Storage

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Analytics Engine

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Interactive Dashboard

Data pipeline transforming raw telemetry into analytics insights

Tech stack

- Python 3.9
- FastAPI
- Pandas
- SQLite + SQLAlchemy
- Chart.js

Key Insights from Telemetry Data

- Peak usage occurs between 14:00–16:00
- Claude Sonnet generates the majority of token usage
- Engineering teams produce the highest request volume
- Average session uses ~600 tokens
- Several usage spikes appear during peak development hours

Interactive Analytics Dashboard

Overview of Claude Code usage patterns

Features:

- KPI metrics
- Usage trends
- Model distribution
- Team activity
- Automated insights

Conclusion

This platform transforms raw telemetry data into actionable insights about developer workflows

It reveals usage patterns, model adoption and resource consumption

Future improvements:

- real-time telemetry streaming
- predictive analytics
- anomaly detection for cost spikes

Built with Python, FastAPI, Pandas, SQLite and Chart.js